

Build a simple DC Motor Driver

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Build a simple DC Motor Driver Circuit using LT-Spice

Part – 1 Circuit Simulation using LT-Spice

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Aim

Very often a DC motor needs to be driven in both forward and reverse directions with controlled operation in embedded and power electronic systems. This build exercise is to learn to design and simulate a simple DC motor driver using an H-bridge circuit, using LT-Spice simulation in an open-loop configuration.

LT-Spice - Introduction

LT-Spice is an open source, high-performance simulation tool developed by Analog Devices. It is used to simulate, design, and test electronic circuits before physically building them.

• Why LT-Spice is important?

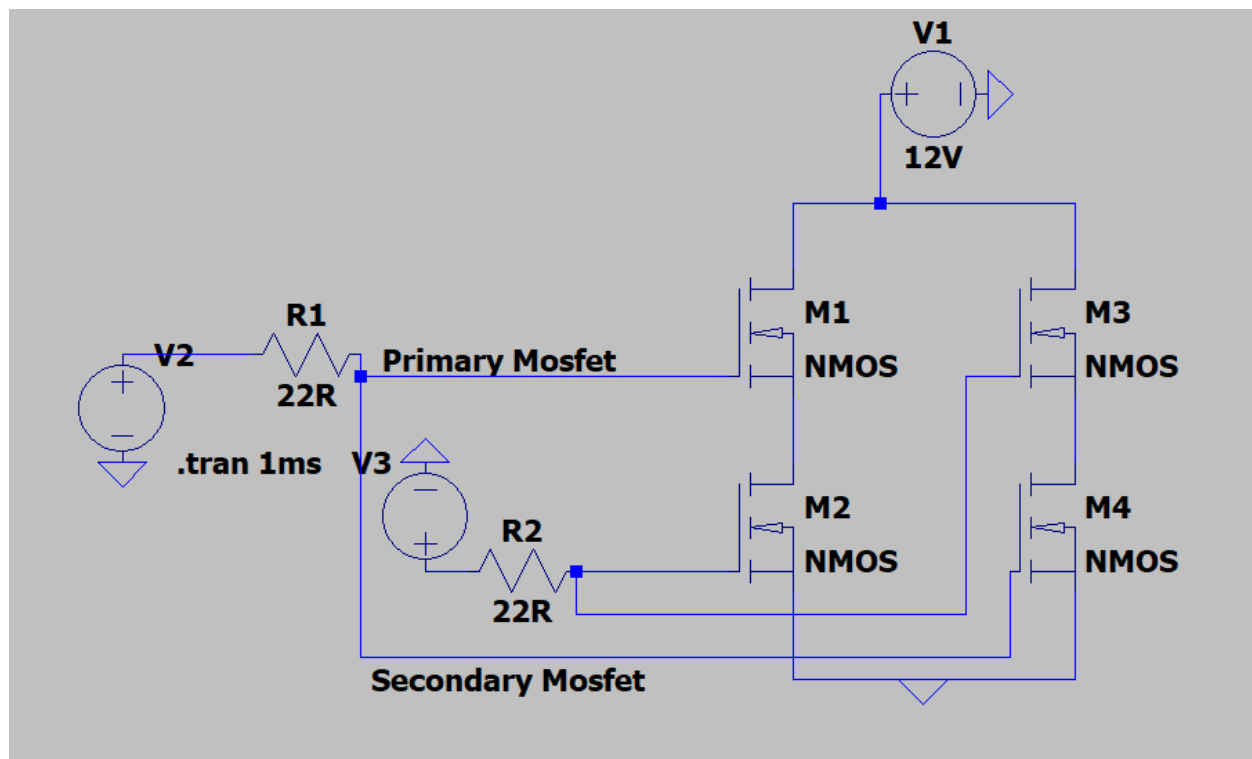
LT-Spice provides a safe learning environment where peoples can freely experiment with electronic circuits without risk of damaging physical components. It helps them to understand circuit behaviour by allowing real time visualization of voltage, current, and other parameters. By simulating before building, Peoples save time and cost as they as they do not need to buy multiple components or PCBs initially. LT-Spice also encourages peoples to develop strong problem - solving skills, teaching them how to identify and fix circuit issues logically. It offers valuable hands-on simulation experience that connects theoretical concepts learned in classrooms with practical real-world electronics. Most importantly, learning LT-Spice gives peoples and industry-relevant skill, preparing them for careers in research, development, and engineering industries.

LT-Spice Simulation Installation Instructions

- 1) Open your web browser (Such as Chrome, Firefox, or Edge).
- 2) In the search bar, type LT-Spice and press Enter.
- 3) Click on the official link from Analog devices, usually titled something like “LTspice” | Analog devices”. Or use this hyperlink to download [LT-Spice](#).
- 4) On the LTspice page, look for a Download button or section.
- 5) Choose the correct version for your operating system (Windows, macOS, etc.).
- 6) Download the installer file to your computer.
- 7) Once the download is complete, open the installer file.
- 8) Follow the on-screen instructions to install LT-Spice:
 - Accept the License Agreement
 - Choose the installation folder (or use the default)
 - Complete the installation process.
- 9) After installation, open LT-Spice from your desktop or Start Menu.
- 10) Now are you ready to start using LT-Spice for circuit simulation.

Connections

Circuit Diagram:



Detailed Connection Steps

Step 1: Open the LT-Spice simulator in your laptop.

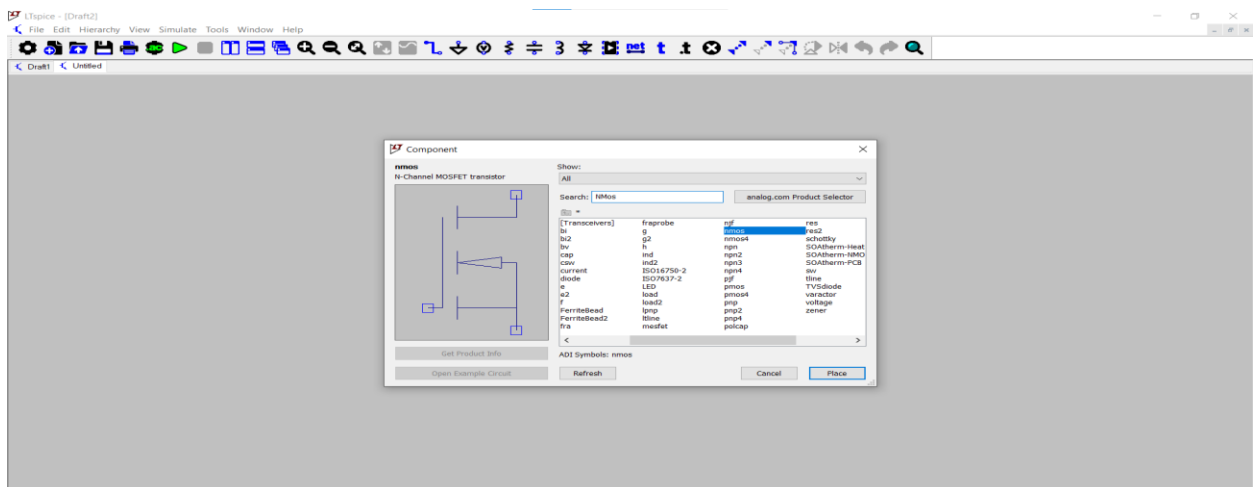
Step 2: Create a new schematic

- Go to the **File** menu.
- Click on **New Schematic**.

Step 3: Place components

We need to place these components one by one:

Component	Symbol in LT-Spice	How to Place
Voltage Source	Voltage(V)	Press F2 then search for voltage
Mosfet	Nmos	Press F2 then search for Nmos
Resistor	Resistor (R)	Press F2, then search for Resistor
Ground	Gnd	Press F2, then search for Gnd



Note: After selecting a component, click on the schematic area to place it.

Step 4: Place Two Voltage Sources

- V1: (Input 12V DC Supply)
- V2 & V3: (PWM gate signal for MOSFET)

Note: In the physical hardware setup, a microcontroller is used to generate the PWM signal required for motor control. The controller produces a PWM output with adjustable duty cycle to control the operation of the H-bridge. In the simulation, the PWM signal is generated using an ideal voltage source, which represents the controller's PWM output and allows the switching behavior to be studied easily.

- After placing the voltage sources, you can right-click on them to rename:
- Name the first voltage source as V1 this will be the input DC supply (e.g., 12V).
- Name the second voltage source as V2 & V3 this will be the PWM control source for the MOSFET gate.

Step 5: Arrange Components Properly

Arrange the components based on the circuit:

- **V1 (DC supply – 12 V)**
 - Positive terminal of V1 connects to the **drains of the high-side MOSFETs (M1 and M3)**.
 - Negative terminal of V1 connects to **ground (GND)**.
- **Left leg of H-bridge**
 - Drain (D) of **M1 (high-side MOSFET)** connects to V1.
 - Source (S) of M1 connects to the **left terminal of the DC motor (R3)**.
 - Drain (D) of **M2 (low-side MOSFET)** connects to the same left motor.
 - Source (S) of M2 connects to **ground (GND)**.
- **Right leg of H-bridge**
 - Drain (D) of **M3 (high-side MOSFET)** connects to V1.
 - Source (S) of M3 connects to the **right terminal of the DC motor (R3)**.
 - Drain (D) of **M4 (low-side MOSFET)** connects to the same right motor terminal.

- Source (S) of M4 connects to **ground (GND)**.
- **DC Motor (Load – R3)**
 - One terminal connects to the midpoint of M1–M2.
 - The other terminal connects to the midpoint of M3–M4.
- **Gate drive connections**
 - Gate (G) of **M1 and M3 (high-side MOSFETs)** connect to the **high-side PWM source (V2)** through resistor **R1**.
 - Gate (G) of **M2 and M4 (low-side MOSFETs)** connect to the **low-side PWM source (V3)** through resistor **R2**.

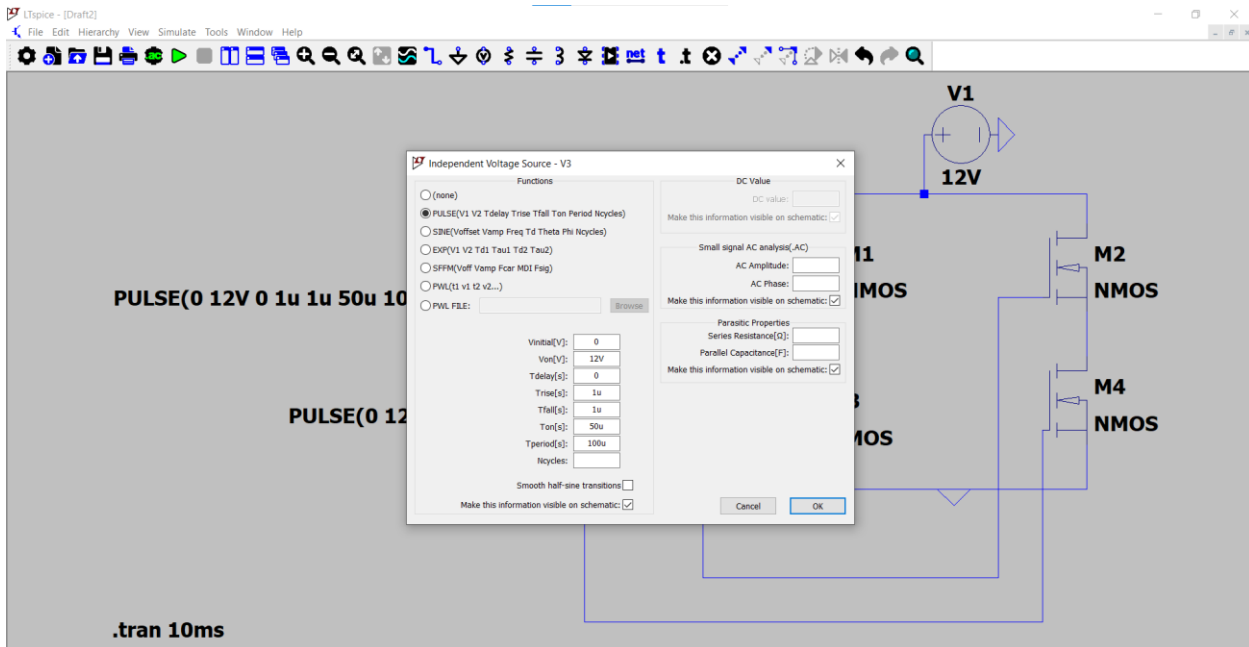
Step 6: Connect Wires

- Press **F3** to start wiring in LTspice.
- Connect each MOSFET terminal exactly as per the H-bridge diagram.
- Ensure all **ground (GND)** symbols are connected to a common reference node.
- Double-check gate connections to avoid shorting high-side and low-side MOSFETs simultaneously.

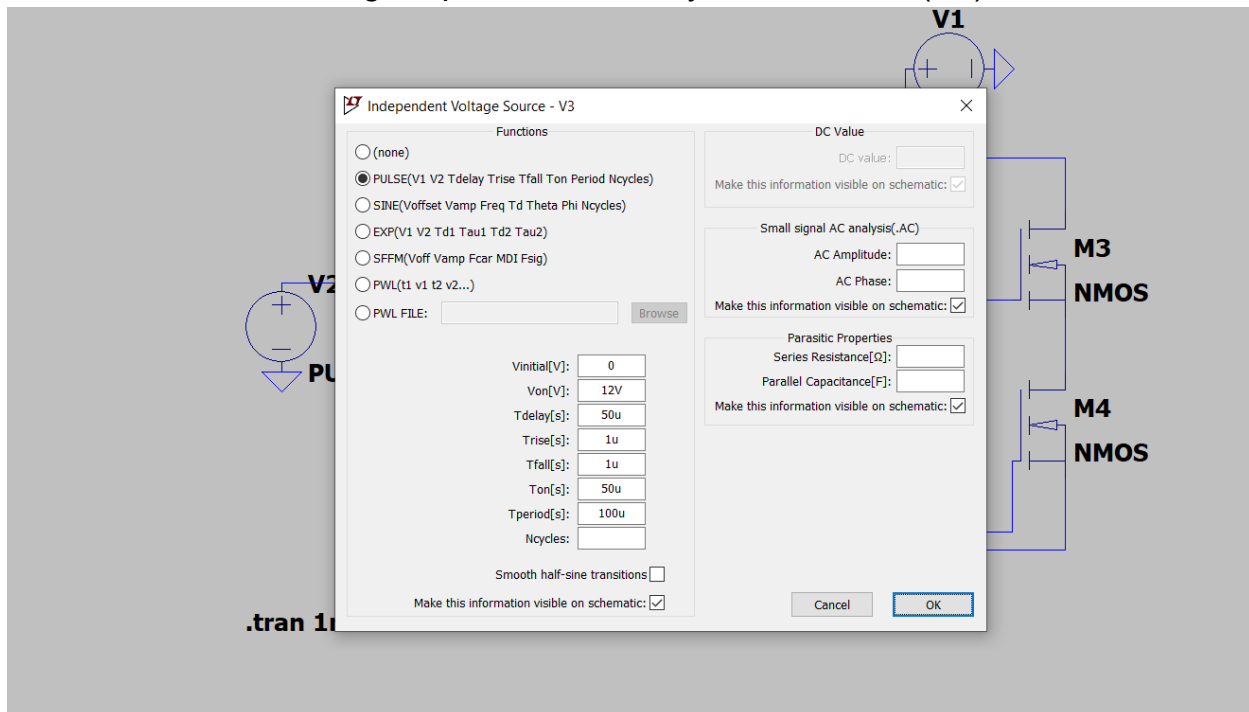
Step 7: Component Values

Component	value
V1(Voltage Source)	DC = 12V
V2 & V3 (PWM Source)	Will set as pulse source (explained below)
Resistor (R1, R2)	22 Ω
Resistor (R3)	1000 Ω (1K)

Note: This will generate a PWM square wave of about 10KHz frequency



Note: The above image represents Primary side of PWM (V2)



Note: The above image represents Secondary side of PWM (V3)

Understanding about Duty Cycle:

The Duty Cycle (%) tells us how long the MOSFET stays ON during one complete switching cycle.

Duty Cycle Formula:

$$\text{Duty Cycle(\%)} = \left(\frac{T_{on}}{T_{period}} \right) \times 100$$

Note: If you want to increase the Duty Cycle then increase the Ton period and observe the output.

Step 8: Add Ground Connections

Every voltage source and component must have a ground reference

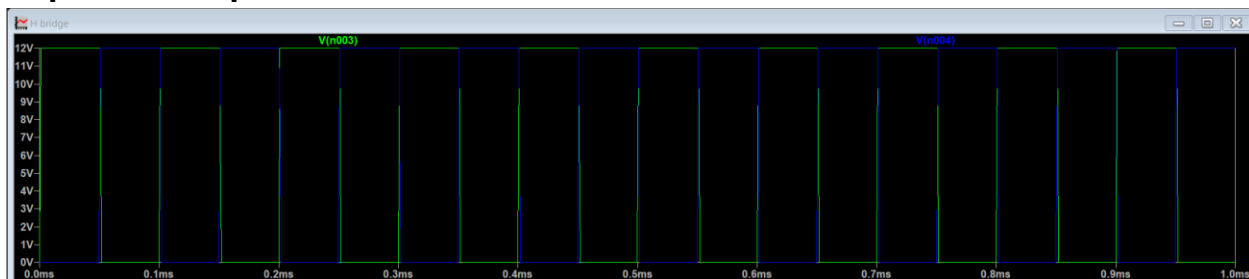
Step 9: Setup Simulation Command

- Press S or go to simulate > Edit simulation Cmd.
- Choose Transient tab.
- Set stop time:
 - 1) Stop time: 10ms
- Click Ok.
- Place the. tran 10ms uic command on the schematic.

Step 10: Save and Run Simulation

- Click save.
- Then click the Run button.
- The simulation will start.
- You can probe:
 - 1) Across MOSFET gate signal (M1).
 - 2) Across MOSFET gate signal (M3).
 - 3) Across the load (R3).

Expected Output:



Note: The reference output waveform demonstrates the H-bridge operation: for each motor terminal, only one MOSFET is ON at a time. When the high-side MOSFET on the positive side is ON (making that terminal HIGH), the low-side MOSFET on the opposite terminal is ON (making that terminal LOW). This complementary switching allows current to flow through the motor in one direction, while preventing both MOSFETs on the same terminal from turning ON simultaneously.

Task:

1. Set up the H-bridge circuit to drive a DC motor with a supply voltage of 12 V. Apply PWM to **one low-side MOSFET** to control the motor speed.
2. Simulate the H-bridge with different **PWM duty cycles** (e.g., 30%, 50%, 70%) and observe how the motor terminal voltage and current change.
3. Change the **load on the motor** (e.g., by increasing motor resistance or adding a series resistor) and observe the effect on motor speed and current.
4. Increase or decrease the **PWM switching frequency** and note how it affects the motor voltage ripple, current waveform, and average speed.
5. Plot and observe the following during simulation:
 - Motor terminal voltage
 - Motor current
 - Gate-to-Source voltage of the PWM-controlled MOSFET

Build a simple DC motor driver circuit using physical circuit

Part 2 – Physical design of the circuit

Aim

Components

Connections - Circuit Diagram

Detailed Steps - Launching Code

Task - Exercise

Aim

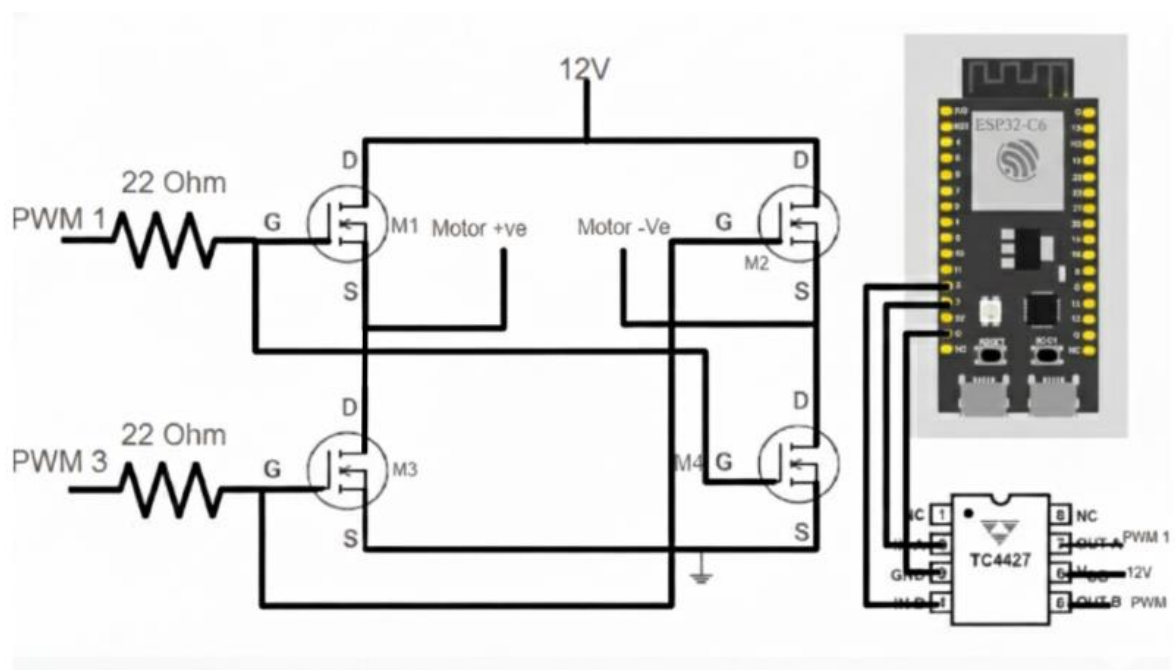
Very often a DC motor needs to be driven in both forward and reverse directions with controlled operation in embedded and power electronic systems. This build exercise is to learn to design and simulate a simple DC motor driver using an H-bridge circuit, using Physical setup in an open-loop configuration.

Components

1. N-Channel logic level Mosfet - 4
2. Resistor (22 ohm) - 4
3. 12V DC Motor - 1
4. Breadboard - 1
5. DC Regulated Power supply or SMPS - 1
6. ESP32C6 Dev Module - 1
7. Jumper Wires - As needed
8. USB to Type - C cable - 1

Note: All the components are reusable.

Circuit Diagram



Detailed Connection Steps

Step 1: Put Parts on the Board

- Place **TC4427 driver** near the MOSFETs.
- Arrange **4 N-channel MOSFETs** in an H-bridge:
 - M1 & M2 → top (high side)
 - M3 & M4 → bottom (low side)
- Put **22Ω resistors** near the MOSFET gates.
- Connect the **motor in the middle**.
- Keep the **ESP32C6 close**.

Step 2: Connect Power

- **+12V** → Drain of M1 & M2 and TC4427 VDD.
- **GND** → Source of M3 & M4, TC4427 GND, ESP32 GND.
 - Make sure all grounds are connected together.

Step 3: Connect the Motor

- Motor **+** → Source of M1 and Drain of M3
- Motor **-** → Source of M2 and Drain of M4

Step 4: Connect MOSFET Gates

- **M1 & M2 gates** → TC4427 OUT_A
- **M3 & M4 gates** → TC4427 OUT_B
- Use **22Ω resistors** for M1 & M3 gates.

Step 5: Connect ESP32 PWM

- ESP32 **GPIO 3** → TC4427 IN_A
- ESP32 **GPIO 2** → TC4427 IN_B
- ESP32 **GND** → common ground

Step 6: TC4427 Enable

- Make sure TC4427 is **enabled**.

Step 7: Check Everything

- Make sure **MOSFETs are oriented correctly**.
- No shorts between **+12V and GND**.
- Motor wires are secure.
- PWM pins in ESP32C6 code match wiring.

Step 8: Power Up

- Power **ESP32** (USB or 5V).
- Apply **12V** to the H-bridge.
- MOSFETs should stay cool.
- Upload **PWM code** to ESP32.

Step 9: Test Motor

- PWM on **IN_A only** → motor goes forward
- PWM on **IN_B only** → motor goes backward
- Change PWM speed → motor changes speed
- Both LOW → motor stops

Full Code:

Update the code soon!

Tasks:

1. Directional Verification

Apply a HIGH signal to TC4427 Input A and LOW to Input B. Record the motor direction. Swap the signals (LOW to A, HIGH to B) and verify the motor spins the opposite way.

2. Speed Regulation via PWM

Apply a PWM signal to Input A (with Input B LOW) at 25%, 50%, and 75% duty cycles. Use a multimeter to measure the average voltage across the motor terminals to see how speed correlates with duty cycle.

3. Power Dissipation Check

Run the motor under its intended load for several minutes. Measure the temperature of the MOSFETs. If they are very hot, use an oscilloscope to measure the voltage drop while they are ON and calculate the power loss.

4. Switching Analysis

Use an oscilloscope to view the voltage at the motor terminals during PWM operation. Look for "ringing" or inductive spikes when the motor turns off. Discuss the role of internal MOSFET body diodes in protecting the circuit.

